ANDERSON VIANA CATALDO

Full-Stack Developer | Automation & RPA | AI Agents (LLM) | Python · TypeScript

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Professional Summary

Full-Stack and Automation Developer focused on **applied AI in software engineering**. I architect and deliver corporate web applications (Next.js 16, React 19, Node.js, FastAPI, PostgreSQL/Supabase) and Python integration robots connected to **TOTVS WinThor ERP (Oracle)**. My key differentiator is orchestrating **AI agents (Claude Code, MCP, Gemini API)** in a *maker-checker* multi-agent architecture, combining **Spec-Driven Development (OpenSpec)** and automated test suites (58 Pytest suites, 35 Playwright assertions, CI/CD) to deliver at maximum speed without compromising code quality, strict typing, or LGPD data privacy. **15 corporate automations delivered**, slashing workflows from hours to seconds (–99%). Computer Science undergraduate at UNIFOR (6th semester), hold an international **Huawei HCIA-Storage V5.0** certification, with advanced English.

Technical Skills

- **AI & Agents:** AI Agents, Multi-Agent Systems, LLM Orchestration, Prompt Engineering, Generative AI, Corrective RAG (CRAG), Function Calling, Evals (95/95), Claude Code, MCP (Model Context Protocol), Google Gemini API, OpenAI API, Guardrails, Fallback & Anti-hallucination, AI-Assisted Engineering.
- **Automation & RPA:** Python, Selenium, Playwright, PyAutoGUI, Web Scraping, ETL, Systems Integration, EDI (Pharma Link), FTP/SFTP (Paramiko, ftplib, WinSCP), Pandas, Tkinter, PyInstaller, Windows Task Scheduler, n8n.
- **Back-end:** Node.js, Express, TypeScript, Python, FastAPI, SQLAlchemy, Alembic, Pydantic v2, Zod, RESTful APIs, JWT, SSO, RBAC; also Java (Spring Boot) and Django.
- **Front-end:** React 19, Next.js 16 (App Router, Server Actions), Vite, Tailwind CSS 4, Recharts, Design Systems, Vue.js, HTML5, CSS3, WCAG 2.1 AA.
- **Data & Storage:** PostgreSQL, Supabase (Auth, RLS, Storage), Oracle SQL (TOTVS WinThor), SQLite, MongoDB, Huawei Enterprise Storage (SAN/NAS/RAID), Google Sheets API / Apps Script, Streamlit, Power BI.
- **Quality, Cloud & DevOps:** Pytest (58 suites), Playwright (E2E & visual regression), Automated Testing, GitHub Actions (CI/CD), Git Flow, Monorepo, Docker, Cloudflare Pages, Render, Vercel, Neon, LGPD Compliance, AppSec (JWT, RLS, CSP, IDOR).
- **Methods & Practices:** Scrum/Agile, Spec-Driven Development (OpenSpec), Maker-Checker Architecture, BDD, Technical Documentation, Figma, Trello, Miro.

Professional Experience

IT Intern — Distribuidora Mais Saúde | Jun/2026 – Present (Fortaleza, CE · Hybrid)

- Architect and build **Central Grupo Mais**, a corporate monorepo with **SSO (JWT)** unifying 4 internal systems (Central Portal, Service Desk, Returns & Reverse Logistics, and IT Asset Management): frontends in Next.js 16/React 19 and Vite, APIs in Node.js/Express (TypeScript) and Python/FastAPI, PostgreSQL (Supabase) with Row-Level Security, deployed on Cloudflare Pages and Render.
- Implemented a **squad of 9 AI agents** (orchestrator, 5 specialized implementers, and 3 independent validators) following the *maker-checker* model with Claude Code: executors never validate their own work, enforcing least privilege, anti-hallucination protocols, and human escalation.
- Turned software quality into an automated delivery gate: **58 Pytest suites** (E2E flows, authentication, and LGPD) and **35 Playwright visual regression assertions**, linting and design system drift checks across 5 packages — 100% approved in production go-live audit.
- Developed **10 Python automations** integrated with WinThor ERP (Oracle), including 4 inventory integration robots for major pharmaceutical operators (Eurofarma, Eurofarma Generic, Smartped, and Momenta) in EDI Pharma Link layout via FTP/SFTP, bundled with PyInstaller.
- Created **FTP_SYNC**, an autonomous bridge between partner FTP servers and WinThor Cloud SFTP (TOTVS) syncing orders, returns, and fiscal invoices every 15 minutes; automated ingestion and ERP loading slashed a 6-hour manual process down to under 10 seconds (–99%).
- Engineered an **AI diagnostic engine** in the Returns System: executive summaries with LLMs (Gemini/OpenAI), Pareto curves of root causes, and anomaly detection with local heuristic fallback (< 1s), inference timeout, and LGPD data anonymization.
- Delivered strategic internal support tooling: Streamlit automation control panel to monitor robot executions, IT asset inventory control (Google Sheets + Apps Script), and Tkinter inventory updater with batch Oracle SQL queries.
- Infrastructure support: corporate networking, hardware provisioning, and level-2 IT help desk.

Apprentice — Tax Department (Systems & Projects) — Grupo 3corações | Jan/2025 – Apr/2026

- Developed **5 Python RPA robots** for automated renewal of tax clearance certificates (CNDs) across government web portals, bulk fiscal XML data extraction, and automated file manipulation.
- Processed more than **15,000 XML files monthly** with 99.85% accuracy against official SEFAZ government schemas.
- Automated web portal navigation and corporate spreadsheet generation from extracted data, eliminating manual rework and cutting over **200 cumulative hours of manual labor**.
- Authored in-depth technical documentation, contingency playbooks, and handled agile ticketing for operational teams.

Competitive Edge: Applied AI in Software Engineering

- **Agent-Driven Development:** Daily practice orchestrating Claude Code with specialized subagents, MCP servers, and strict delegation workflows to plan, implement, review, and audit code in production systems.
- **Spec-Driven Development:** Formal OpenSpec/BDD specifications authored before code; autonomous agents query and update specifications across iterations, preserving context and eliminating hallucinations.
- **Automated AI Quality & Security Gates:** Autonomous QA and AppSec validators run builds, Pytest, and Playwright suites, auditing JWT, RLS, CSP, and IDOR vulnerabilities before approving pull requests.
- **First-Principles Mastery:** Built an autonomous multi-agent squad from scratch (Node.js + Gemini API + PM2) with role personas, shared JSON memory bus, and 24/7 supervisor daemon.
- **Responsible AI in Production:** Integration of LLMs with business systems enforcing strict Zod/Pydantic output schemas, deterministic fallbacks, and LGPD data anonymization.

Featured Projects

- **Central Grupo Mais & Corporate SSO (Corporate):** Unified corporate monorepo with JWT SSO integrating 4 mission-critical platforms (Central, Tickets, Returns, Assets). Built with Next.js 16, React 19, Vite, Node.js, FastAPI, Supabase (RLS), Cloudflare Pages + Render.
- **Corporate Service Desk System — Mais Saúde (Corporate):** Multi-tier service platform (requester → agent → manager → admin) with granular RBAC, predictive SLA tracking, MTTR analytics, interactive Recharts BI, batch actions, and automated PDF reporting. Next.js 16, React 19, TypeScript, Supabase (PostgreSQL, RLS), Tailwind CSS 4.
- **AI-Powered Reverse Logistics & Returns (Corporate):** Replaced Google Forms/spreadsheets: triage, duplicate invoice validation, warehouse TV Mode, real-time alerts, and executive AI diagnosis (Gemini/OpenAI) with Pareto analysis and sub-second fallback. React + Vite, Python/FastAPI, SQLAlchemy, Alembic, Pydantic v2, Pytest.
- **FTP_SYNC & WinThor EDI Engine (Corporate):** Autonomous SFTP bridge connecting pharmaceutical logistics partners (Eurofarma, Smartped, Momenta) in EDI Pharma Link layout with TOTVS WinThor ERP (Oracle), slashing a 6-hour workflow to under 10 seconds (–99%). Python, SFTP/FTP, Paramiko, PyInstaller.
- **IT Asset Management & SHA-256 Custody (Corporate):** Hardware inventory tracking with 2-tier approval flow, database audit triggers, and automated .docx Responsibility Term generation signed with SHA-256 cryptographic hashes. Next.js, Node.js, PostgreSQL (Supabase).
- **Autonomous AI Agent Squad (Applied Research):** Multi-agent architecture built from scratch: role personas, shared JSON memory bus, finite state machine orchestrator, and PM2 daemon for continuous 24/7 operation. Node.js, Google Gemini API, PM2.
- **Gamified Educational Platform — São Carlo Acutis (Personal):** Full-stack web application featuring interactive quizzes, real-time scoring, and educational gamification. React, TypeScript, Tailwind CSS, Supabase, Vercel.
- **SaaS Web Portfolio & Anderson AI Agent (Personal):** Full-stack showcase built to Silicon Valley craft standards with global Command Palette (Ctrl+K), executable CLI, RPA ROI simulator, lazy-loaded AI assistant (hybrid 5-stage CRAG + CoVe, 95/95 evals, Gemini 3.8/3.5 with Function Calling, L1/L2 Supabase cache with negative-feedback invalidation, AI_WRITE_SECRET RPCs, LGPD purge), WCAG 2.1 AA, and PWA. React, TypeScript, Tailwind CSS 4, Vite 6, Vercel Edge, Supabase.

Education & Certifications

- **Bachelor of Science in Computer Science:** University of Fortaleza (UNIFOR) | 2024 – In progress (6th semester).
- **Huawei HCIA-Storage V5.0:** Huawei ICT Academy | Certified Jun/2026 (Enterprise Storage, SAN/NAS, RAID, Disaster Recovery).

Languages & Behavioral Competencies

- **Languages:** Portuguese (Native) · English (Advanced) · Spanish (Basic).
- **Cross-Functional Communication:** Fluent collaboration with non-technical business teams, translating operational pain points into automated software solutions.
- **Proactivity & Ownership:** Autonomous identification of manual bottlenecks and rapid execution of high-leverage software pipelines.
- **Agility & Discipline:** High performance in fast-paced environments with concurrent demands, clean code standards, and strict delivery gates.